Tutorial - 07

Q) Customer Purchase Prediction

An E-comerce platform wants to predict whether a customer will make a purchase using KNN.

Input features:

- → Time on site (minutes)
- → pages viewed
- → previous purchases.
- → cart value (dollars)

Target variable:

- → purchase (Yes = Y, NO = N)

Training dataset

CustomerID	Timeonsite	pageswand	pages purchased	cost value	Purchase
1	25	8	5	120	Y
2	5	2	0	0	N
3	30	10	7	100	Y
4	8	3	1	20	N
5	22	7	4	100	Y
6	6	2	0	15	N
7	28	9	6	140	Y
8	7	3	1	25	N

Q)

1) calculate euclidean distance for Test instance T_1 and classify using $k = 1, 3, 5$

2) calculate classification accuracy for $k = 1, 3, 5$ which k performs best?

3) create confusion matrix for $k = 3$?

4) Calculate precision and Recall for "Purchase=Y" class when $k = 3$.

Test Dataset (with Actual labels)

TestID	Time on site	Page viewed	previous Purchases	Cost value	Actual purchase
T1	20	6	3	90	Y
T2	10	4	2	40	N
T3	21	9	5	130	Y

Sol $T_1 = (20, 6, 3, 90)$

formula =

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 + (w_1 - w_2)^2}$$

customer	values	Distance from T_1	purchase
1	(25, 8, 5, 120)	$\sqrt{(25-20)^2 + (8-6)^2 + (5-3)^2 + (120-40)^2} = 30.54$	Y
2	(5, 2, 0, 0)	$\sqrt{(5-20)^2 + (2-6)^2 + (0-3)^2 + (0-90)^2} = 91.37$	N
3	(30, 10, 7, 150)	$\sqrt{(30-20)^2 + (10-6)^2 + (7-3)^2 + (150-90)^2} = 61.09$	Y
4	(8, 3, 1, 20)	$\sqrt{(8-20)^2 + (3-6)^2 + (1-3)^2 + (20-90)^2} = 71.111$	N
5	(22, 7, 4, 100)	$\sqrt{(22-20)^2 + (7-6)^2 + (0-3)^2 + (15-90)^2} = 10.29$	Y
6	(6, 2, 9, 15)	$\sqrt{(6-20)^2 + (2-6)^2 + (0-3)^2 + (15-90)^2} = 76.45$	N
7	(28, 9, 6, 140)	$\sqrt{(28-20)^2 + (9-6)^2 + (6-3)^2 + (140-90)^2} = 50.81$	Y
8	(7, 3, 1, 25)	$\sqrt{(7-20)^2 + (3-6)^2 + (1-3)^2 + (25-90)^2}$	N

Sorted Nearest Neighbours

Rank	customer	Distance	Class
1	5	10.30	Y
2	1	30.56	Y
3	7	50.82	Y
4	3	61.12	Y
5	8	66.39	N

Classification of T1 :-

when K=1, Nearest Neighbour customers (Y)

Prediction = Y

when K=3, Nearest Neighbours

customers (5, 1, 7) → (Y, Y, Y)

prediction = Y

when K=5, Nearest Neighbours

customers (5, 1, 7, 3, 8) → (Y, Y, Y, Y, N)

Majority = Y

prediction = Y

2) Classification Accuracy

Actual labels vs predicted labels

Test	Actual	predicted	predicted K=1	predicted K=3	predicted K=5
T1	Y	Y	Y	Y	Y
T2	N	N	N	N	N
T3	Y	Y	Y	Y	Y

An predictions match with the actual labels

. Accuracy = $\frac{\text{correct predictions}}{\text{Total predictions}}$

3) Confusion Matrix

	predicted Y	predicted N
predicted N	True positive	False Negative
Actual N	False positive	True Negative

	PY	PN
Actual Y	2	0
Actual N	0	1

TP = 2

FN = 0

FP = 0

TN = 1

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{2+0} = 1 = 100\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+0} = 1 = 100\%$$